MOISEL RARES

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ABOUT ME

I'm passionate about graphics and physics programming, with a particular interest for C++ especially its meta-template programming features. I greatly enjoy the architectural aspects of software development.

EDUCATION

Master Studies in Graphics, Multimedia, and Virtual Reality
University POLITEHNICA Bucharest, Bucharest, Romania 2025-2027

- *Related coursework*: Graphics pipeline processing programming, Development of Virtual and Augmented Reality Systems, Parallel Programming for GPGPU, etc.
- *Thesis:* In progress.

Bachelor Studies in Mathematics and Informatics
University of Bucharest, Bucharest, Romania 2022-2025

- *Related coursework*: Major in Informatics with minor courses in Mathematics
- *Thesis:* Construction of Polygonal Surfaces from Signed Distance Fields

PROJECTS

Algorithms for Retrieving Meshes from implicit surfaces
In this project, I implement and optimize algorithms like Marching Cubes, Surface Nets, and Dual Contouring to extract polygonal meshes from implicit surfaces, converting complex shapes into 3D models for rendering and editing.

DirectX11 Graphical Engine
A graphical engine developed using DirectX 11, focused on good architecture for enhanced dynamic control, designed to render 3D graphics. Written in **C++**.

Terrain Rendering
Developed real-time terrain rendering using procedural heightmaps with accurate normal map generation. Implemented tessellation shaders for efficient LOD and applied multiple texturing techniques. Written in **C++**.

Dungeon Crawler Game
Developed a dungeon crawler in **Unity** with procedural map generation, combat and item upgrades.

Other Projects
Built small projects including an OpenGL graphics engine, raymarching, Bezier curves, edge detection, and a horror game in **UE5**.

SKILLS AND TECHNOLOGIES

Programming languages and concepts
C/C++, C#, Python, Bash, CMake, Git, Object-Oriented Programming (OOP), multithreading and concurrency, GPU programming (GPGPU), graphics programming (OpenGL/Vulkan), Shader programming (GLSL/HLSL), Linux/ Unix environments, cross-platform development, Unreal Engine 5 (UE5), Unity Engine, VR/AR, Linear Algebra, Computational Geometry, 3D Math.

REFERENCES

References available upon request.